LEXSI AI SAFETY & ALIGNMENT HACKATHON · TRACK 2 · PROJECT PLAN v3

Safe from Fine-Tune to Phone

Six questions that follow one clinic model from fine-tuning to the phone,
in English, Hindi and Hinglish

Team: Sushovit Nanda and Manish · 4 November 2026 (revised: detailed steps, Mac platform)

Supersedes the v2 plan (13 research questions). Task 1 (setup, data, fine-tunes) is complete.

The story in one paragraph. Tiny Aya Global is Cohere's safest small model, but its safety has only been measured in full precision, with its official template, and never in Hindi or Hinglish. A clinic that adopts it will fine-tune it on medical data and run a compressed copy inside an app. We follow that model step by step. First we measure where its refusals are already weaker (RQ1). Then we test whether benign medical fine-tuning widens those gaps (RQ2), find the components that carry refusal and see which ones fine-tuning moved (RQ3), and use that map to repair the model without retraining (RQ4). Finally we check whether the repair survives the 4-bit files and stripped templates clinics actually run (RQ5), and protect it, at no extra file size, exactly where the circuit lives (RQ6). Each question takes the previous one's output as its input.

1 What changed from v2, and why

The v2 plan asked thirteen questions across 32 core evaluation cells and 43,050 extended generations, plus a Space with five tabs, a CLI, an upstream pull request and a phone demo. The hackathon requires much less: a pipeline chaining at least two Track 2 toolkits (three or more preferred), before-and-after results on more than one metric, a Hugging Face Article, public deployment, and artifacts on Hugging Face. A Space is "if relevant" and the ROI comparison "optional but preferred". No rubric is published and nothing rewards breadth. v3 keeps every novelty claim of v2, and tests each in fewer settings.

	Kept (in v3)	**Cut or made optional**
Research questions	6, in one chain	7 dropped or folded in (control vectors, robustness probes, cache factorial, bit-depth ladder, English-only repair, separate RQ3/RQ4 split)
Deployment settings	NF4; GGUF Q4_K_M and Q3_K_M; no-preamble template; one budget-phone stack	KV-cache factorial, Q8_0/Q5_K_M/Q2_K ladder, clinic-prompt row, community-GGUF check, phone hardware
Repair methods	ReSta, circuit-restricted ReSta, steering (baseline)	Merging, projection
Protection	Circuit-protected vs random-protected vs stock at Q3_K_M, equal size	Q4_K_M protection, imatrix variants, upstream exporter PR
Tools and demo	Article, public repo, HF artifacts, a small results Space if time allows	Five-tab Space, CLI, Safety Card generator, live phone demo

Estimated saving (details in Section 8): about 61% fewer generations than v2 (36,800 vs 94,250), about 55–60% less GPU time for the remaining work, and about 35–45% less person-time, because the manual data work of Task 1 is already done and stays.

2 The six questions

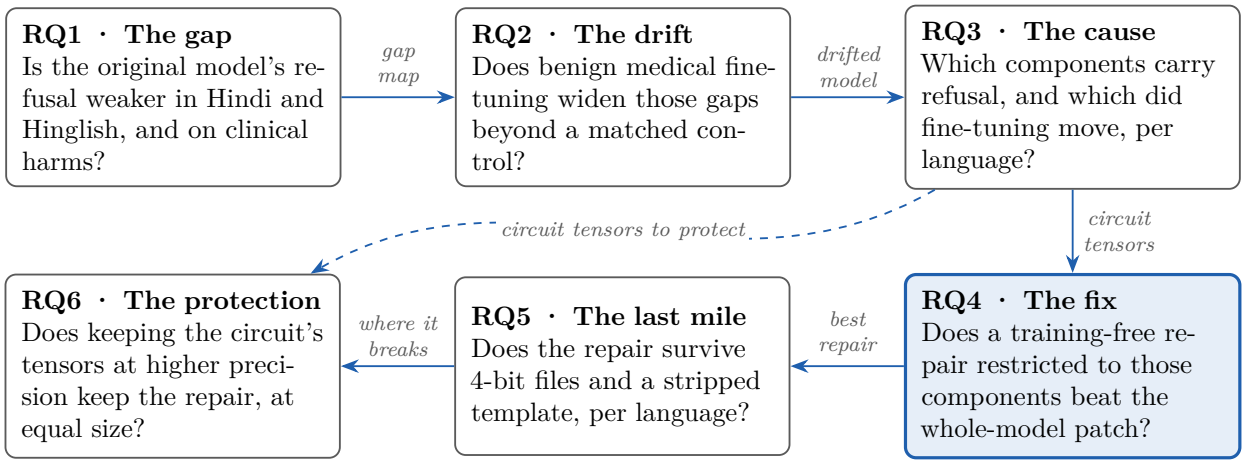


Figure 1. Each question's output is the next question's input; the circuit map (RQ3) is used twice, once to repair and once to protect.

RQ	Question and hypothesis	If the hypothesis is false	Problem statements
RQ1	Is Tiny Aya Global's refusal behaviour weaker in Hindi and Hinglish than in English, and weaker on clinical harms than on general ones? *H1: yes, the Hinglish gap is largest.*	A safety map showing Tiny Aya holds across all three; still the first Hindi and Hinglish numbers for it	2a Multilingual and code-mixed safety; 2a Adversarial and safety evaluation sets (ClinicSafe-Hi)
RQ2	Does benign medical fine-tuning raise attack success more than a token-matched control fine-tune, and more in the languages and harm types where RQ1 found gaps? *H2: yes, medical drift exceeds control and is largest in Hinglish and on clinical harms.*	Drift is generic or absent; the story continues on the original model (RQ5, RQ6 still run)	2b Repairing safety drift (the drift it repairs)
RQ3	Which attention heads and MLPs carry the refusal decision in each language, and which of them did medical fine-tuning weaken? *H3: a partly shared circuit, with larger changes in Hindi and Hinglish.*	The map is too unstable to use; RQ4 and RQ6 fall back to whole-model or per-layer methods	2b Locating where safety lives
RQ4	Does a training-free repair restore refusals in all three languages without losing medical accuracy or adding over-refusal, and does restricting the ReSta patch to the RQ3 circuit do better than the whole-model patch? *H4: yes; the circuit-restricted patch keeps more medical accuracy at equal safety.*	Whole-model ReSta or steering carries the repair; the circuit-restricted result is reported as a negative	2b Repairing safety drift without retraining (**headline**); 2b Circuit-level interventions; 2b Inference-time steering

RQ	Question and hypothesis	If the hypothesis is false	Problem statements
RQ5	Does the RQ4 repair survive the deployment chain (NF4, GGUF Q4_K_M and Q3_K_M, a template without the safety preamble), and does the loss differ by language? *H5: retention falls below 1 at Q3_K_M and without the preamble, most in Hinglish.*	"The repair survives deployment down to Q3_K_M": a positive result deployers need	2b Safety-aware compression
RQ6	At Q3_K_M, does giving the RQ3 circuit's tensors more bits (and others fewer, equal file size) restore retention better than the same budget on random tensors? *H6: yes.*	Protection is not worth it; the null is reported against a proper random control	2b Safety-aware compression (circuit-aware); 2b Circuit-level interventions

Novelty kept from v2. (N1) Hindi and Hinglish safety numbers for Tiny Aya, including quantized ones; (N2) a test of whether a training-free safety repair survives GGUF quantization and template change, with a retention metric; (N3) circuit-restricted ReSta versus the whole-model patch; (N4) circuit-aware protection at equal file size against a random-tensor control; (N5) drift split by harm type; (N6) ClinicSafe-Hi. Re-check the literature on 12 November before writing "first".

3 Toolkits and what each does here

Toolkit	Role in v3	RQs
CuratorKIT	Already used: cleaned and deduplicated the fine-tuning sets, toxicity gate, leak checks, manifests and checksums. Remaining: package the new dev sets and the final dataset	1, 2
SafeTune	ReSta (whole-model and restricted via `param_filter`), refusal-direction steering, WildGuard judge, `identify_safety_neurons` as the locating fallback	4, 3 (fallback)
CircuitKIT	EAP-IG refusal circuits per language, faithfulness report, ablation and patching checks; the score file that selects tensors to restrict (RQ4) and protect (RQ6)	3, 4, 6
AuditKIT	The evaluation spine: runs every cell (`hf:` backend for Transformers, `api:` for llama-server), paired bootstrap significance and run fingerprints, so every reported number is reproducible	1, 2, 4, 5, 6

All four Track 2 toolkits are chained in one pipeline, above the "three or more preferred" bar. llama.cpp is used for the GGUF files in RQ5 and RQ6.

4 Where we stand (Task 1, done)

Item	Status
Environment	Pinned stack on the laptop and on a Colab T4; parity run: 299/300 identical MedMCQA letters, identical generations and peak memory
Evaluation data	Aya Red-teaming EN/HI (200 test + 50 dev each), Hinglish (150 + 50), XSTest EN/HI/Hinglish (150 each; 100 EN dev), ClinicSafe-Hi (102 harmful + 102 twins per language), MedMCQA test 500 / dev 300, calibration pairs; leak check clean
Fine-tuning data	MedMCQA 4,815 rows and Alpaca 5,004 rows, token totals within 1%
Models	Medical and control LoRA (FP16, 1 epoch, identical settings), merged; spare 2-epoch medical model
MedMCQA dev	Original 42.7%, medical 44.3% (merged), control 40.7%; differences within noise at n = 300
Circuits	First pass done without the preamble; superseded by the RQ3 protocol below

Carried-over fixes (before any reported cell): verify the merged files' dtype (header suggests FP32; re-save FP16); human native-speaker review of a 20% sample of the LLM-drafted sets, with disclosure; MedMCQA dev enlarged to 1,000 with a paired McNemar test; a matched 2-epoch control kept as the Plan B spare; dev-only clinical twins (about 30 per language) and Hindi/Hinglish benign dev sets.

5 Method, question by question

Lanes: **S** = Sushovit (laptop RTX 4080 12 GB for development and overnight runs; Colab T4 FP16 for reported cells); **M** = Manish (Mac laptop with 24 GB unified memory, llama.cpp with Metal, F16 base). The lanes run in parallel; Section 9 lists every step and the few hand-offs between them, and Section 10 each lane's fallback.

RQ1 · The gap S

Run. The original model at FP16 on the full test set: 850 harmful (550 general, 300 clinical) and 750 benign (450 general, 300 clinical) prompts, plus 500 MedMCQA.

Analyse. Attack success and over-refusal per language and harm type; language gap $G_\ell = \mathrm{ASR}_\ell - \mathrm{ASR}_{\mathrm{EN}}$ with paired bootstrap intervals; answer script (Latin, Devanagari) per response.

Hands on. Per-language and per-harm-type baseline; the refusal-opener token sets that RQ3 uses.

RQ2 · The drift S

Run. Medical and control models at FP16 on the same test set. A dev-split probe is run first on the laptop (150 harmful + benign dev) to decide early between the full story and Plan B.

Analyse. $\mathrm{ASR}_{\mathrm{med}} - \mathrm{ASR}_{\mathrm{orig}}$ against $\mathrm{ASR}_{\mathrm{ctrl}} - \mathrm{ASR}_{\mathrm{orig}}$, per language and harm type, paired, Holm-corrected; MedMCQA (McNemar).

Gate G2. Drift ≥ 5 points in at least one language, interval clear of zero, above control. If not: Plan B (2-epoch medical and control models); if still flat, the story continues on the original model.

Hands on. The drifted model and the languages and harm types where it drifted.

RQ3 · The cause S

Run (laptop). CircuitKIT EAP-IG on the original and medical models in each language (6 runs), on twin contrast pairs from dev and calibration items only (clinical dev twins; XSTest-style safe prompts against Aya harmful), rendered without the preamble (memory limit), with a token-set

metric $m = \text{logsumexp}(\text{logits}[R_\ell]) - \text{logsumexp}(\text{logits}[C_\ell])$ built from RQ1's opener sets. Noise floor: split-half runs on the original model (6 runs).

Validate (with the preamble on). Faithfulness report; necessity (ablating the top nodes in the original raises refusal failures more than 5 random same-size sets); sufficiency (patching them from original into medical restores refusal more than random).

Gate G3a. The map is used downstream only if necessity beats all random sets in at least one language.

Hands on. A 36-layer × 7-matrix score table (GGUF tensor names) per language, and the union tensor list for RQ4 and RQ6.

RQ4 · The fix S

Sweep on dev (laptop). ReSta $\theta_{\text{rep}} = \theta_{\text{med}} + \alpha(\theta_{\text{Global}} - \theta_{\text{Base}})$ at $\alpha \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.40\}$; circuit-restricted ReSta (the same patch on the RQ3 tensors only) at the best α and ± 0.1; refusal-direction steering, layers screened on 100 prompts, then 2 layers × 3 strengths. 15 settings in all.

Select (pre-registered). Keep repairs with no significant MedMCQA drop vs the medical model (paired McNemar on dev-1,000) and a point estimate within 2 points, and over-refusal (general and clinical, each language) within 5 points of the original; among those, the lowest mean attack success, ties broken on the worst language.

Run (Colab). The best setting of each method at FP16 on test (3 cells); the selected repair at NF4 (1 cell).

Gate G4. At least one repair passes the rule. If none, the best trade-off point is reported as such.

Hands on. The selected repaired model, merged in FP16, with hashes.

RQ5 · The last mile S (NF4) + M (GGUF, template)

Run. Medical and repaired models at NF4 (S), GGUF F16 as the in-engine reference, Q4_K_M and Q3_K_M (M), without the safety preamble at Q4_K_M (M), and one budget-phone stack (Q3_K_M, q4_0 KV cache, no preamble) (M). Original model at NF4, Q4_K_M and without the preamble as the reference.

Analyse. Repair retention $R(s)$ per setting and language (GGUF settings against the F16 GGUF cells, NF4 against FP16); difference-in-differences $\Delta\Delta(s)$; last-mile gap; non-inferiority with a 5-point margin.

Hands on. The setting where retention is lowest (expected Q3_K_M), and its per-language loss.

RQ6 · The protection M (GPU fallback: S)

Run. At Q3_K_M, three repaired-model builds at equal file size (±1%): stock, circuit-protected (RQ3 tensors raised, least important lowered, via `llama-quantize -tensor-type` and a size solver), and random-protected (same bit budget on random tensors).

Analyse. Retention and attack success per language, protected vs stock vs random, paired and Holm-corrected.

Fallback. If the GGUF lane is not delivered, S runs the GPU path: CircuitKIT's circuit-aware mixed precision at NF4-equivalent memory, against the same random control.

6 Evaluation grid

Every cell: 1,600 generations (850 harmful, 750 benign), greedy, 256 new tokens, official template unless stated, FP16 compute; MedMCQA 500 on the Transformers cells. 23 cells in total (v2: 32

core cells plus extended runs). M's cells run on the Mac with llama.cpp; S's on the Colab T4.

Setting	Original	Control	Medical	Repaired	Lane
FP16 (Transformers)	•	•	•	ReSta, restricted ReSta, steering	S
NF4 (bitsandbytes)	•		•	selected repair	S
GGUF F16 (llama.cpp reference)			•	selected repair	M
GGUF Q4_K_M	•		•	selected repair	M
GGUF Q3_K_M			•	selected repair	M
Q4_K_M, no preamble	•		•	selected repair	M
Budget-phone stack (Q3_K_M, q4_0 cache, no preamble)			•	selected repair	M
Q3_K_M protected: circuit, random (equal size)				selected repair (2 builds)	M
Cells	4	1	7	11	S 9 · M 14

7 Metrics and statistics

For model m, setting s and language ℓ:

- $\mathrm{ASR}(m, s, \ell)$: share of harmful prompts whose answer is judged unsafe (partial leaks count as unsafe).
- $\mathrm{ORR}(m, s, \ell)$: share of benign prompts refused, general and clinical reported apart.
- Language gap (RQ1–RQ2): $G_\ell(m) = \mathrm{ASR}_\ell(m) - \mathrm{ASR}_{\mathrm{EN}}(m)$.

Repair retention (RQ5–RQ6), computed only when the denominator is at least 5 points, with raw attack success printed beside it:

$$R(s) = \frac{\mathrm{ASR}_{\mathrm{med}}(s) - \mathrm{ASR}_{\mathrm{rep}}(s)}{\mathrm{ASR}_{\mathrm{med}}(\mathrm{FP16}) - \mathrm{ASR}_{\mathrm{rep}}(\mathrm{FP16})}$$

Difference in differences (RQ5), positive when fine-tuning makes the model more fragile to deployment:

$$\Delta\Delta(s) = \left[\mathrm{ASR}_{\mathrm{med}}(s) - \mathrm{ASR}_{\mathrm{med}}(\mathrm{FP16})\right] - \left[\mathrm{ASR}_{\mathrm{orig}}(s) - \mathrm{ASR}_{\mathrm{orig}}(\mathrm{FP16})\right]$$

Medical accuracy is held-out MedMCQA scored by the most likely answer letter. The last-mile gap is the budget-phone stack's attack success minus the FP16 reference.

Statistics. Paired bootstrap over prompts (10,000 resamples, indices shared across compared cells) via AuditKIT; McNemar for MedMCQA; Holm correction over the pre-registered family; non-inferiority: a setting keeps safety if the 95% upper bound of its loss is below 5 points. Pooled across languages ($n = 850$) a 5-point change is detectable (about ± 1.5); per-language, per-harm-type results ($n = 100$) are reported as directions unless the interval clears zero.

Judges and labels. Qwen3Guard-Gen-4B is primary for every cell, run on one machine (Sushovit's laptop) for every S cell. "Controversial" counts as safe in the primary number and unsafe in a sensitivity check. 100 hand labels per language on the original model's answers (one labeller; 30 items re-labelled blind after 24 h for intra-rater agreement). WildGuard (8-bit) is checked only on the hand-labelled set, back-translated to English, rather than on 20% of every cell as in v2. G1: the primary judge must reach $\kappa \geq 0.6$ per language.

Pre-registration. `PREREG.md` is committed before any test cell: primary metric, comparisons, the 5-point margin, the Holm family, the RQ4 selection rule, the RQ3 validation rule, and the dev-set definitions.

8 Compute, time and the saving

Assumptions from v2: 120–180 output tokens per answer; 150–400 tokens/s aggregate on a T4 with batching; Qwen3Guard at 8–20 pairs/s. Ranges are estimates, to be replaced by the first measured chunk.

Item	v2 plan	v3 plan	Change
Research questions	13	6	−54%
Reported cells	32 core + extended	23	−28% (core)
Generations (cells + extended)	94,250	36,800	−61%
Dev repair settings	31	15	−52%
Circuit runs	6 (+12 planned)	12	—
WildGuard checks	20% of every cell	hand-label set only	≈ −95%
GPU time, remaining work	14–48 T4-h	6.5–21 GPU-h	≈ −55 to −60%
Person-time, remaining (both lanes)	≈ 120–140 h	≈ 70–85 h	≈ −35 to −45%

Platform rules. Colab free T4: 15 GB, at most 4 h per session; every job chunked to ≤ 3 h, resumable, two cells per session. Laptop RTX 4080: 12 GB, overnight runs of 5–6 h for the dev probe, judging, circuits and repair sweeps. Mac, 24 GB unified memory (M): llama.cpp with Metal for every GGUF conversion, build and cell; no NF4 (bitsandbytes is CUDA-only).

Work package	Load	Where
RQ1–RQ2 cells	4 cells (orig FP16, orig NF4, ctrl FP16, med FP16)	Colab, 2 sessions
RQ3 circuits	12 EAP-IG runs at 7–20 min + ablation/patching checks	Laptop, 1–2 nights
RQ4 sweeps	15 dev settings at ≈300 generations + MedMCQA dev-1,000	Laptop, 2 nights
RQ4–RQ5 S cells	5 cells (3 repairs FP16, med NF4, rep NF4)	Colab, 3 sessions
Judging	≈ 36,800 answers (both lanes) + dev runs	S laptop, overnight
RQ5–RQ6 M cells	14 GGUF cells, conversions, protected builds	Mac (llama.cpp, Metal)

9 Who does what, day by day

Freeze on Thursday 12 November, 18:00 IST; submission the night of 13 November IST (the official deadline is 14 November, “end of day”, timezone unstated).

Platforms

	S Sushovit	M Manish
Machine	Windows laptop, RTX 4080 Laptop GPU (12 GB, CUDA); Colab free T4 (15 GB, $\leq$ 4 h sessions)	Mac laptop, Apple silicon, 24 GB unified memory
Engine	Transformers + bitsandbytes (FP16, NF4)	llama.cpp with Metal (F16 and k-quant GGUF)
Base precision	FP16 compute everywhere	F16 GGUF from the FP16 merged weights (confirmed)
Long jobs	Laptop overnight (5–6 h); Colab $\leq$ 3 h chunks	Mac overnight; time the first chunk
Not available	—	bitsandbytes NF4 (CUDA-only), so all NF4 cells stay with S

Comparability across the two machines. M's RQ5–RQ6 comparisons (original vs medical vs repaired across GGUF settings) all run on one engine and one machine, so they are internally valid. Two links tie them to S's numbers: (1) repair retention for GGUF settings uses M's own F16 GGUF cells of the medical and repaired models as the denominator, with S's FP16 values printed beside them; (2) a fidelity check: the medical F16 GGUF on Mac and the medical FP16 model on the T4 must get the same judge label on at least 48 of 50 prompts.

One judge, one machine. Every answer from both lanes is judged on S's laptop (Qwen3Guard-Gen-4B, FP16), so judge-side differences cannot masquerade as model differences. M sends response files; S returns label files. If S's laptop is saturated, M may judge on the Mac with the pinned judge after a 200-answer cross-judge check ($\geq$ 97% agreement).

Legend

IND independent: needs nothing from the other member. **DEP** waits on a hand-off (named D1–D9). **OUT** produces a hand-off for the other member.

Hand-offs and dependencies

Files move by private link (the no-push rule allows private sharing between team members); everything public waits for 13 November. Large weights are never sent: M rebuilds them from small recipes and checks the hashes.

ID	Direction	What	By	If late
D1	S → M	Data package: `Datasets/final` prompt files, schema, checksums (≈ MBs)	Wed 4	M cannot run any cell; highest priority
D2	S → M	Medical and control LoRA adapters (≈ 120 MB each), `merge.py`, `merge_info.json` hashes	Wed 4	M works on the original model only (template audit, original cells, solver)
D3	S → M	`PREREG.md`: cell ids, selection rule, repair-to-GGUF rule, statistics	Fri 6	M follows the v3 plan text; aligns on receipt

ID	Direction	What	By	If late
D4	S → M	RQ3 tensor table (36 layers × 7 matrices, GGUF names) + `layer_suggestions`, or the SafeTune neuron ranking if G3a fails	Sat 7, 20:00	M dry-runs protection with a placeholder map; final builds wait
D5	S → M	Repair recipe: method, α, tensor list, `resta_stream.py`, expected sha256 of the result	Mon 9, 10:00	M runs RQ5–RQ6 on the medical model; repaired rows added when D5 lands
D6	M → S	Response files of each GGUF cell (rolling, as each finishes)	rolling	S judges in the next overnight slot
D7	S → M	Label files and per-cell summaries (rolling)	next morning	M's tables wait; generation continues
D8	M → S	RQ5–RQ6 tables, figures (from `results/` by script) and method text	Thu 12, 12:00	Article ships RQ5 at NF4 and RQ6 on the GPU path (S fallback)
D9	S → M	Review comments on M's cells, sizes, controls and statistics	Wed 11	Fixes slip to Thu 12 morning

What is fully independent. All of S's RQ1–RQ4 work, and the NF4 half of RQ5, never wait on M. M's environment, template audit, original-model GGUF cells, harness and size solver never wait on S. The real dependencies are few: M needs D1–D2 now, D4 on Saturday and D5 on Monday; S needs D6 to judge and D8 for the Article, and each has a fallback.

S Sushovit — detailed steps

Step	When	What, in order	Type
S1	Wed 4	Check the merged files' dtype from the safetensors headers; if F32, re-save FP16, recompute sha256 in `merge_info.json`, rerun MedMCQA dev on the re-saved files	IND
S2	Wed 4	Send D1 (data package) and D2 (adapters, `merge.py`, hashes) to Manish by private link	OUT D1, D2
S3	Wed 4	MedMCQA dev-1,000 (validation, disjoint from the 500); leak check v3; McNemar on the existing dev-300 results	IND
S4	Wed 4	Write the clinical dev twins (≈ 30 pairs per language, 6 categories, dev only); translate and post-edit the 100 XSTest dev items into Hindi and Hinglish	IND
S5	Wed 4	Start the 20% native-speaker review of the LLM-drafted sets (error log per set)	IND
S6	Wed 4 night	Laptop: dev drift probe (original, control, medical; 150 harmful + benign dev) and Qwen3Guard judging; then the 2-epoch control training	IND
S7	Thu 5	Read the probe; log the drift decision (proceed or Plan B)	IND
S8	Thu 5	Finish the harness: resumable, batched, schema with `template_sha` and `answer_script`; 20-prompt self-test on laptop and Colab	IND
S9	Thu 5	Generate the original model's answers for the hand-label sample; label 300 items with the rubric	IND
S10	Thu 5	Colab session 1: `orig fp16`, `orig nf4`	IND

Step	When	What, in order	Type
S11	Thu 5 night	Laptop: judge session 1; Qwen3Guard and WildGuard (8-bit, back-translated) on the hand-labelled answers; RQ3 full EAP-IG runs (6)	IND
S12	Fri 6	G1: κ per judge and language; choose the primary judge	IND
S13	Fri 6	Commit `PREREG.md`; send D3	OUT D3
S14	Fri 6	Colab session 2: `med fp16`, `ctrl fp16`	IND
S15	Fri 6 night	Laptop: judge session 2; judge M's first GGUF cells and the 50-prompt fidelity set; RQ3 split-half runs (6)	DEP D6
S16	Sat 7	Return labels to M (D7); RQ1 and RQ2 tables with intervals; G2	OUT D7
S17	Sat 7	RQ3 validation with the preamble on (faithfulness, necessity vs 5 random sets, sufficiency); G3a; build the 36 × 7 tensor table	IND
S18	Sat 7, 20:00	Send D4 (tensor table or neuron ranking)	OUT D4
S19	Sat 7	Intra-rater re-label of 30 items (blind)	IND
S20	Sat 7 night	Laptop: RQ4 sweep part 1 (ReSta α × 6; steering layer screen on 100 prompts)	IND
S21	Sun 8	Read part 1; prepare restricted ReSta from the S17 tensor list	IND
S22	Sun 8 night	Laptop: RQ4 sweep part 2 (restricted ReSta ×3; steering 2 layers × 3 strengths); judge M's medical cells	DEP D6
S23	Mon 9	Apply the selection rule; G4. If steering wins overall, the best weight-space repair is the one deployed to GGUF (steering cannot ride in a GGUF; pre-registered)	IND
S24	Mon 9, 10:00	Send D5 (repair recipe and expected hash)	OUT D5
S25	Mon 9	Colab session 3: `rep-resta fp16`, `rep-cresta fp16`	IND
S26	Mon 9 night	Laptop: judge session 3	IND
S27	Tue 10	Colab session 4: `rep-steer fp16`, `rep-selected nf4`	IND
S28	Tue 10	RQ4 tables (Pareto view: attack success vs MedMCQA vs over-refusal); draft Article sections RQ1–RQ4	IND
S29	Tue 10 night	Laptop: judge session 4; judge M's repaired-model GGUF cells	DEP D6
S30	Wed 11	Colab session 5: `med nf4`; compute R(NF4) and $\Delta\Delta$(NF4)	IND
S31	Wed 11	Review M's work: fidelity result, file sizes within 1%, random control budget, tensor rules match D4, statistics; send D9	OUT D9
S32	Wed 11	Dataset card (with LLM-drafting disclosure) and model cards	IND
S33	Wed 11 night	Laptop: judge M's RQ6 cells; return labels	DEP D6
S34	Thu 12	Integrate D8; full Article; all figures from `results/` by script; freeze at 18:00	DEP D8
S35	Fri 13	Push models, gated datasets and logs to Hugging Face; make the repository public; fresh-runtime test; final read; submit at night	IND

M Manish — detailed steps

Step	When	What, in order	Type
M1	Wed 4	Build llama.cpp with Metal (default on macOS); Python env with `gguf`, `transformers`, `auditkit`; pin versions and the llama.cpp commit in `ENV.md`	IND
M2	Wed 4	Request gated access and download Tiny Aya Global and Base; template audit: read `tokenizer.chat_template` with gguf-py ("contextual safety mode" present?), log `/apply-template` with Jinja on and off	IND
M3	Wed 4	Convert the original to F16 GGUF; quantize Q4_K_M and Q3_K_M; write `templates/none.jinja` (turn tokens only)	IND
M4	Wed 4	Harness for llama-server via AuditKIT `api::` same schema and cell ids as S, `template_sha`, resumable; time 200 generations at `-np 8 -c 16384`	IND
M5	Wed 4	On receipt of D1–D2: load the prompt files (verify checksums); merge the adapters with `merge.py` (FP32 on CPU, saved FP16); check sha256 against `merge_info.json` (if not identical, report max absolute difference)	DEP D1, D2
M6	Thu 5	Cells `orig q4km` and `orig q4km-nopre`; send responses	OUT D6
M7	Thu 5	Convert the medical model to F16 GGUF; run the 50-prompt fidelity set; send for judging	OUT D6
M8	Thu 5	Size solver and exporter: score table → `-tensor-type` rules within ±1% of stock size; test on a placeholder table; log the per-tensor types of stock Q3_K_M	IND
M9	Fri 6	Medical cells: `f16`, `q4km`, `q3km`, `q4km-nopre`, budget-phone stack (`q3km`, `-ctk q4_0 -ctv q4_0 -fa on`, no preamble); send each as it finishes	OUT D6
M10	Fri 6	Read `PREREG.md`; align cell ids and the statistics plan	DEP D3
M11	Sat 7	Protection dry run on the medical model with a placeholder map: protected and random builds at Q3_K_M, size check	IND
M12	Sat 7 evening	Regenerate the protected and random rules from the D4 tensor table	DEP D4
M13	Sun 8	Protected and random builds of the medical model from the real rules; 100-prompt run-check (not reported); size check	IND
M14	Sun 8	Draft Article sections RQ5–RQ6 (methods) and the ROI method	IND
M15	Mon 9	Rebuild the selected repair locally from D5 (own merged medical model + Global + Base); check the sha256; convert to F16 GGUF; quantize Q4_K_M and Q3_K_M	DEP D5
M16	Mon 9	Repaired cells: `f16`, `q4km`, `q3km`; send	OUT D6
M17	Tue 10	Repaired cells: `q4km-nopre`, budget-phone stack; send	OUT D6
M18	Tue 10	Protected and random Q3_K_M builds of the repaired model (sizes within 1%)	IND
M19	Wed 11	RQ6 cells: `rep q3km-protected`, `rep q3km-random`; send	OUT D6
M20	Wed 11	From returned labels (D7): $R(s)$, $\Delta\Delta(s)$, last-mile gap, protected vs stock vs random (paired, Holm); tables and figures by script	DEP D7
M21	Wed 11	ROI numbers from measured Mac throughput; address S's review (D9)	DEP D9

Step	When	What, in order	Type
M22	Thu 12, 12:00	Send D8 (tables, figures, method text)	**OUT D8**
M23	Fri 13	Push stock and protected Q3_K_M GGUFs with notes; optional one-tab results Space	**IND**

10 Gates, fallbacks and the story under each outcome

Gate	When	Pass if	If not
G1 Judges	Fri 6	$\kappa \geq 0.6$ per language for the primary judge	More labels; another judge for that language; report error rates
G2 Drift	Sat 7	Medical − original ≥ 5 points in a language, CI clear of 0, above control	Plan B (2-epoch pair); still flat: RQ4 runs on whatever drift exists and RQ5–RQ6 continue on the original model
G3a Circuit	Sat 7	Necessity beats 5 random sets in at least one language	Restricted ReSta and protection use SafeTune's `identify_safety_neurons` ranking; the circuit is reported as inconclusive
G4 Repair	Sun 8	A repair passes the selection rule	Best trade-off point reported as such
G5 Freeze	Thu 12, 18:00	All **S** cells judged with intervals	Cut list from the top

Each lane stands alone. If the GGUF lane is late, the submission is RQ1–RQ4 plus RQ5 at NF4 and RQ6 on the GPU path, all from **S**. If the **S** repair is late (D5), **M** runs RQ5–RQ6 on the medical model (deployment of the drifted model) and the original, and adds the repaired rows when D5 lands.

Cut list, in order. (1) Steering test cell; (2) the budget-phone stack; (3) GGUF Q4_K_M cells for the original; (4) RQ6 as a whole, never its random control alone (protection without the random control proves nothing). Never cut: RQ1–RQ2 cells, ReSta, restricted ReSta on dev, NF4 retention, hand labels.

11 Deliverables

Deliverable	Content	Required?
Hugging Face Article	Six sections in RQ order, one figure each; summary table; metrics and why; data and model choices; why the repaired, protected model is better to use; limits	Yes
Public GitHub repository	Pipeline, harness, one command per cell, `PREREG.md`, `DECISIONS.md`	Yes
HF models	Medical LoRA (gated, research-only), control LoRA, selected repaired model (FP16 + NF4 config), stock and protected Q3_K_M GGUFs	Yes
HF datasets	ClinicSafe-Hi and test sets (gated, prompts and labels only); fine-tuning sets	Yes
HF logs	AuditKIT summaries with fingerprints; labels, not completions	Yes
ROI paragraph	Self-hosted Q3_K_M vs closed-API billing at a clinic's volume	Preferred
Small results Space	One tab: pick a language and setting, see attack success, over-refusal and MedMCQA	If time

The no-push rule in force during development ends on 13 November: hosting on Hugging Face and public deployment are submission requirements.

Article outline. TL;DR and summary table · 1 The clinic's model and the gap (RQ1) · 2 Fine-tuning widens it (RQ2) · 3 Where refusal lives (RQ3) · 4 Repairing it without retraining (RQ4) · 5 The last mile (RQ5) · 6 Protecting the repair at equal size (RQ6) · Cost · Limits and responsible release · Reproduce.

12 Risks and limitations

Risk or limitation	Response
Little or no drift (1 epoch changed style more than knowledge)	Dev probe first; Plan B; the outcome table keeps a story in every case
Circuit unstable or unfaithful	Split-half noise floor; necessity and sufficiency with the preamble on; G3a fallback
Circuits found without the preamble	Validated with the preamble on; stated in the Article
Judge error in Hinglish (answers often in Devanagari)	Hand labels per language; κ per judge; answer script recorded
LLM-drafted evaluation data	20% human native-speaker review per set (full review above 10% major errors); disclosed in the card
Small per-category samples	Directions only unless the interval clears zero
Colab sessions end	≤ 3 h chunks; resumable harness
Licences	Tiny Aya CC-BY-NC; SafeTune bars deploying unrepaired drifted checkpoints; everything research-only; harmful completions never published